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Project Work

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List of Abbreviations

DVB Dresdner Verkehrsbetriebe

GAMS General algebraic modeling system

List of Symbols

M	a sufficient large constant number. 10
d_{lij}	travel time on line l . 10
f_{ij}	Fixed cost. 3
i, j	edge. 3, 10
l	Installed line. 10
od_{uv}	Passenger demand from origin u to destination v . 3, 10
x_{uiv}	arriving inbound passengers in node V . 3
x_{ujv}	arriving inbound passengers in node V . 3
x_{ulij}	Passenger flow from origin u traveling on line l . 10
x_{uvi}	outbound passengers of node V . 3
x_{uvj}	outbound passengers of node V . 3
y_{ij}	Binary variable, equals 1 if edge (i, j) is installed; 0 otherwise. 3
$y_{l=1}$	if line l is installed (0; otherwise). 10

1 Introduction

This Project Work is addressed to create campus shuttle in Dresden. Nowadays in Dresden there are persistence challenges due to the high demand caused by a large number of Students and staff from TU Dresden. As Dresdner Verkehrsbetriebe (DVB) is responsible for transport operation services in Dresden, experience different challenges such as delays and overcrowding particularly the routes that closely link the University facilities. To handle these challenges, the purpose of this project is to answer the research Question: How can a campus shuttle in Dresden minimize the costs and overall travel time at the same time reducing congestion? Based on these challenges this project is composed with two main tasks: Route Network Planning and Line network planning. The purpose of the first main task is to create the optimal network by expanding current network with existing edges with 30 stops, then increase the stops up to 32 to form a new planned campus shuttle by Using General algebraic modeling system (GAMS) to create the model for new route network and analyse the data respectively to the costs (construction costs and usage costs) and implemented edges in the network with different type of buses (Small, medium and Large). The Purpose of the second task is to form line network by using also GAMS model to minimize travel time and analyse how the travel time and number of Implemented lines will change respectively to the way the actual budget was used. And also to see how the line network will changes when the number of lines to be implemented are limited. The aim of this Project is to improve the mobility in these routes to be more efficiency by minimizing travel times and overcrowding. As this is an expensive project, the main point also is to find the way to minimize the costs such as Usage costs and constructions costs while implementing also the optimal bus type, this will led the project run smoothly and efficiently. The following is analyzing obtained results and explaining the research methodology used in this Project.

2 Route network Planning

2.1 Problem description

The Existing Transport system operated by Dresdner Verkehrsbetriebe DVB is highly used by Student and Staff from TU Dresden, this lead to delay, congestion and overcrowding. That's why the University implemented this project to solve this problem by expanding and modifying where it's necessary campus shuttle in Dresden. This system has to connect the main campus in the region of Südvorstadt, the location in Johannstadt, the Faculty of Medicine, the Botanical Garden and the student residences Dresden. For enabling this project run efficiently, route network design must be implemented for planning campus shuttle. The route has to include the existing 30 stops. Then additionally, we include other 2 stops-(Postplatz (stop 31) and Zwinglistraße (stop 32). We have different scenarios, that we use to analyze the results by considering how the number of installed edges, usage costs and fixed costs varies depending on the considered stops as well as depending on the type of bus(small, medium and large) implemented and its passenger capacity limit. The main goal is to determine number of installed edges for each bus type, passenger origin-related traffic flows, edge loadings and to minimize total costs (Usage costs and construction costs). According to the task description we have to analyze different nine cases. Each case have to be analyzed and compared with other cases based on the number of obtained edges, Total costs (Usage costs and Fixed costs) to identify the most reliable and optimal solution for this implementation.

2.2 Mathematic Model

Objective Function

$$\min G = \sum_{[i,j] \in E} \left(f_{ij} \cdot y_{ij} + c_{ij} \cdot \sum_{u \in V} (x_{uij} + x_{uji}) \right) \quad (2.1)$$

Coupling Constraints

$$x_{uij} + x_{uji} - B_u \cdot y_{ij} \leq 0 \quad \forall u \in V, [i,j] \in E \quad (2.2)$$

Flow Preservation Constraints

$$\sum_{[i,v] \in E} (x_{uiv} - x_{uvi}) - \sum_{[v,j] \in E} (x_{uvj} - x_{ujv}) = od_{uv} \quad \forall u, v \in V, u \neq v \quad (2.3)$$

Capacity Constraints

$$\sum_{u \in V} x_{uij} \leq \text{Bus Capacity} \quad \forall [i, j] \in E \quad (2.4)$$

The mathematical model is used to minimize the total costs of campus shuttle route network related to Bus capacity and passenger demand constraints. The Objective Constraints 2.1 is used to minimize the total costs consisting of fixed costs (construction costs) and opportunity costs (variable costs). f_{ij} means fixed costs that can be implemented once when edge i, j was installed and reciprocal to binary variable $y_{ij}=1$ while i, j means usage costs that can varies due the passenger volume across installed edge (i, j) . Coupling constraints 2.2 is used to ensure that passenger can only be transported from i to j or from j to i , if the edge i, j is installed and the summation of the passenger from i to j and j to i must not exceed node capacity B_u when $y_{ij}=1$. Flow Preservation constraints 2.3 is used to ensure that the difference between arriving and continuing passengers equals to the demand od_{uv} . This avoid the route network to be unbalanced due to a certain routes have more passengers than demand. x_{uiv} and x_{ujv} means arriving inbound passengers in node v while x_{uvi} and x_{uvj} means continuing/outbound passengers of node v . Capacity constraints 2.4 is used to implement the bus type capacity limit (large: 63000, medium: 45000, small: 31000 passengers per day for each direction). These Equations are used in solving 9 different cases consisting of existing route network with 30 stops, 30 stops+ Postplatz and 30 Stops + Postplatz + Zwinglistraße respectively with bus types.

Route Network for Stops 1 to 30				
Type of bus	Total costs	Variable costs	Fixed costs	Number of edges
Large bus	3,619,661.20	3,409,661.20	210,000.00	41
Medium bus	3,812,384.60	3,626,384.60	186,000.00	43
Small bus	4,229,183.40	3,980,183.40	249,000.00	47
Route Network for Stops 1 to 31 (added stop <i>Postplatz</i>)				
Large bus	3,701,684.20	3,513,184.20	188,500.00	42
Medium bus	3,860,668.20	3,663,168.20	197,500.00	46
Small bus	4,398,024.10	4,138,524.10	259,500.00	51
Route Network for Stops 1 to 32 (added stops <i>Postplatz</i> and <i>Zwinglistraße</i>)				
Large bus	3,906,684.20	3,720,684.20	186,000.00	45
Medium bus	4,171,830.00	3,947,830.00	224,000.00	49
Small bus	4,751,716.00	4,500,716.00	251,000.00	53

Table 2.1: Obtained results of routes Networks for nine cases

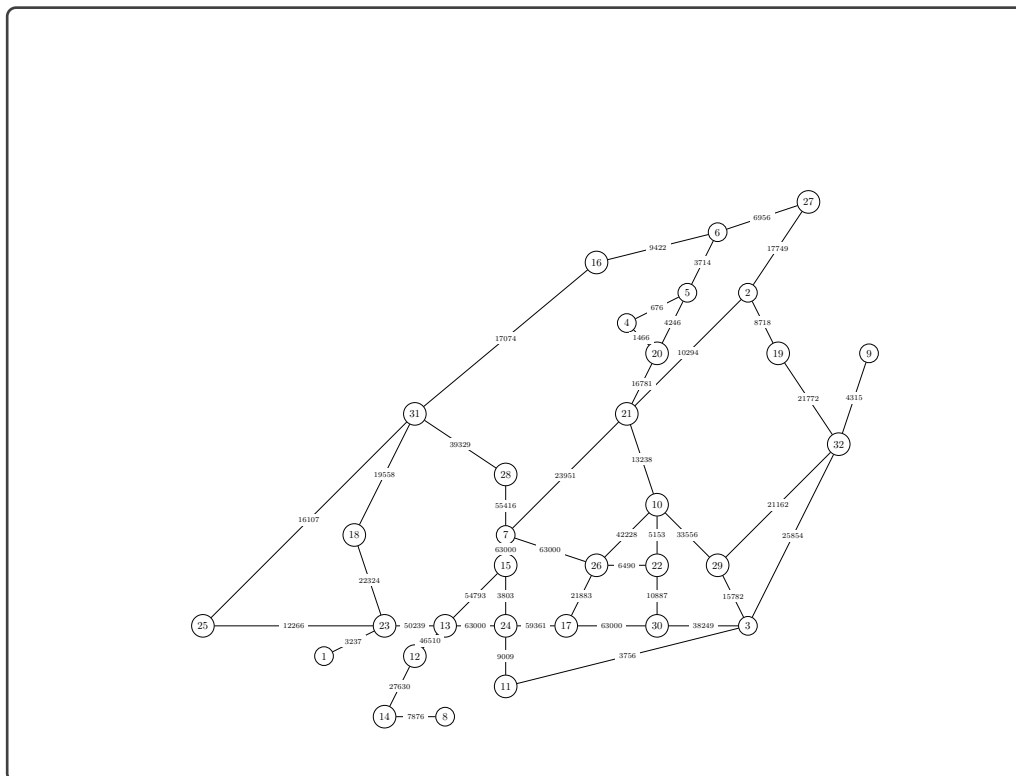


Figure 2.1: a figure with the determined route network when employing the large buses with 32 stops and the edge loading for both directions

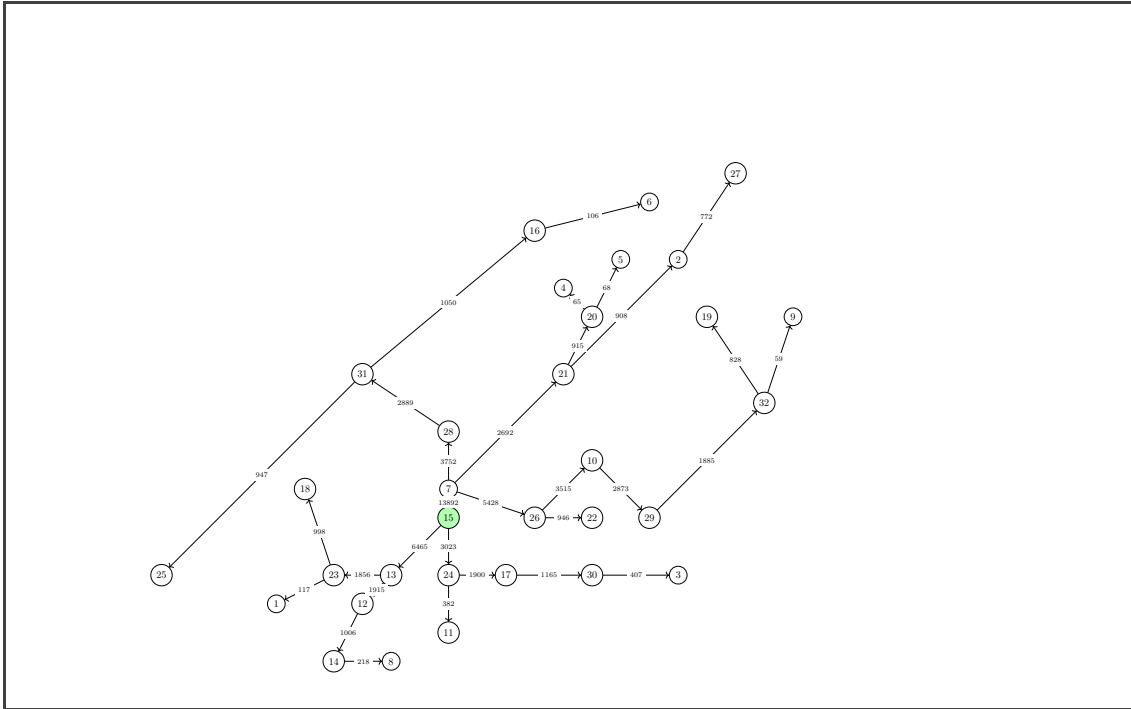


Figure 2.2: a figure with the origin-related traffic flows for the stop Reichenbachstraße (node 15) when employing the large bus, with arcs that shows direction the passengers are traveling.

2.3 Comparison and analysis of obtained Route Network results

This route network of campus shuttle Dresden was created in different 9 cases: existing network (30 stops), existing network + Postplatz(node 31), existing network +Postplatz(node 31)+ Zwinglistraße (node 32) and each case was evaluated with 3 different types of Buses(small, medium, Large). Every route network consists of total costs(fixed costs and variable costs), number of installed edges and Bus types. For Route network with 30 stops, large buses have less total costs (€3,619,661.20) due to the buses with high capacity this also lead to the minimization of variable costs(€3,409,661.20) although causes high fixed costs(€ 210,000) with fewer installed edges(41). The next case is medium bus with total costs(€3,812,384.60) with high variable costs(€3,626,384.60) because of reduction of bus capacity to 45,000 passengers per day but with lower fixed costs (€186,000) and compared with large buses, number of installed edges was increased up to 43 edges. Small bus types have worst results compare to large and medium bus types because it have high variable costs(€3,980,183.4) and fixed costs(€249,000) due to work inefficiency at less capacity (31,000 passengers) , this lead also the increasing of edges up to 47. By extending the route network to 31 stops(adding Postplatz), large bus types remained the most effective with total costs(€3,701,684.20) with low fixed costs(€188,500) and with fewer number of installed edges(42). The next case

of medium buses also is cost effective compared to small bus type as it performed worst with high total costs (€4,398,024.1) with variable costs (€4,138,524.10) due to the reduction of capacity(31,000 passengers) this lead also to increasing the number of edges up to 51. For 32 stops(adding Zwinglistraße), large bus types continue to lead with total costs(€3,906,684.20) with lowest fixed costs(€186,000) and very efficiency number of implemented edges(45). Due to the expanding the route to 32 stops, the demand increase while small bus capacity (31,000 passengers) is less, this causes small bus types continues to underperform with high total costs(€4,751,716) with high usage costs(€4,500,716) due to increasing of demand and high number of edges(53) needs to be installed, but the fixed costs(€251,000) is slightly low compare 31 stops cases on small buses. Variable costs dominated more than 95% of total costs in all cases due to passenger volume while fixed costs are less because it has to be implemented once. According to the results analyzed above, it shows that the addition of Node 31(Postplatz) to Node31 (Zwinglistraße) further continue to increase costs and edges. The large bus type continued to be more efficient compare to the other bus types. 2.1 shows edge loading for both direction when employing large buses for 32 Stops, the edges was increased up to 45 due the expansion of the route network while maintaining flow efficiency without congestion. 2.2 shows how passengers flows for stop 15 (Reichenbachstraße) when employing large buses, the implemented arcs shows the direction of incoming and outgoing passengers while also maintain flow efficiency without overcrowding.

2.4 Results of modified route network of 32 stops

2.4.1 Route network of 32 stops for different bus types with and without edge [7,10]

	Type of bus	Total costs	Variable costs	Fixed costs	Number of edges
Without the edge [7,10]	Large bus	3,906,684.20	3,720,684.20	186,000.00	45
	Medium bus	4,171,830.00	3,947,830.00	224,000.00	49
	Small bus	4,751,716.00	4,500,716.00	251,000.00	53
With the edge [7,10]	Large bus	3,842,839.80	3,668,439.80	179,800.00	43
	Medium bus	4,093,207.60	3,866,407.60	226,800.00	50
	Small bus	4,620,526.80	4,379,726.80	240,800.00	53

Table 2.2: Results of modified route network of 32 stops for different bus types with and without edge [7,10].

Including the edge [7,10] that connect Lennéplatz and Hauptbahnhof with fixed costs of 5,800€ and usage costs of 2.7€ changes the route network, edge loadings and costs (variable costs and fixed costs) across all Bus types. For large buses, the number of edges reduced from 45 to 43 while also all costs reduced, this make the route network becomes more efficiency because this added new edge reduced some long path. For medium buses, number of edges increased from 49 to 50 while costs (variable costs and fixed costs) reduced, this slightly expands the network but also it remains efficiency due to the reduction of costs. For small bus, the route network retains 53 edges that mean another edge was removed to maintain optimal connectivity but the costs (variable costs and fixed costs) reduced. Total costs were decreased for all bus types after including this edge. For Large buses, total costs reduced by €58,444.4. For medium buses, the total costs was decreased by €78,622.4, largely because variable costs decreased by €81,422.4 although fixed costs slightly increased by €28,00. These results shows also that implementing this new edge take up some demand from other edges reduces congestion which enhance network efficiency especially high demand stop like Hauptbahnhof.

2.4.2 Route network by considering toll to reduce traffic jams in some stops

affected edges	Modified usage cost
7 - 15	6.4
7 -18	57.6
7 - 21	28
7 -23	47.2
7 - 26	9.6
7 -28	14.4
10 - 19	60
10 - 21	13.6
10 - 22	8.8
10 -26	18,4
10 - 29	19.2
21 - 2	42.4
21 - 19	60
21 - 20	18.4
21 - 26	40.8
21 - 28	45.6
28 - 16	59.2
28 - 18	54.4
28 - 20	33.6
28 - 31	9.6
31 - 16	23.2
31 - 18	19.2
31 - 25	24

Table 2.3: Calculated Passenger-related usage costs for all affected edges

2 Route network Planning

	Type of bus	Total costs	Variable costs	Fixed costs	Number of edges
Without toll	Large bus	3,906,684.20	3,720,684.20	186,000.00	45
	Medium bus	4,171,830.00	3,947,830.00	224,000.00	49
	Small bus	4,751,716.00	4,500,716.00	251,000.00	53
With toll	Large bus	8,408,229.40	8,169,729.40	238,500.00	47
	Medium bus	8,953,741.20	8,708,741.20	245,000.00	49
	Small bus	10,037,667.80	9,749,167.80	288,500.00	52

Table 2.4: Results of route network with and without toll for all three bus types

Introducing a city center toll, which increases passenger-related usages costs by a factor of 8 for edges connected to Hauptbahnhof (node 7) , Lennéplatz (node 10) , Straßburger Platz (node 21) , Walpurgisstraße (node 28) and Postplatz (node 31), changes the route networks particularly dramatically increasing of Costs(Variable costs and fixed costs) across all bus types. For Large busses, the Total costs increased by 115% caused by nearly doubled of variables costs, as toll was applied to the passenger usage costs, though the number of edges was only slight expanded from 45 to 47 edges. For Medium buses, when a toll was introduced, caused also dramatically increasing of Total costs by 114% due to doubled variable costs due to increasing of passenger related usage costs, but number edges consist a route network was retained, although the total costs was nearly doubled. For Small Buses, total costs were increased by 111% and the number of edges was decreased from 53 to 52. Compared small bus types with large busses and medium busses shows that the costs was not dramatically increased while number of edges decreased due to the low capacity of the busses. These modifications of route network causes the distributions of edges loadings by shifting the traffic from the city centre routes to non-central routes. Although introducing this toll on identified stops on the networks reduces traffic congestion, it increases the operating costs for keeping the network continues to run more efficiently.

3 Line Network Planning

3.1 Problem description

This part of project “Campus Shuttle in Dresden ” is focusing on Line networking generated from route network, based on initial route network include 32 stops and potential connection between stops. This line network planning project is divided in 4 parts: first part is to determine the optimal line network from route network without considering the budget to calculate the overall travel time, arising costs and number of implemented lines. The second part is to modify the line network by expanding the model with the budget constraint and reduce the 10% of operation costs from initial line network that was taken as reference to the available budget and repeat this process until no feasible solution provided. For each model run, Available budget, Overall travel time, actually used budget and Number of implemented line must be recorded, afterwards a scatter plot that shows the dependence travel time and the number of implemented lines from available budget as well as the line network for last feasible solution. Because of infrastructure reasons, the edge [13, 15] have to be closed temporary that will affect the 13 lines due to the replacement of this edge with [13,24] and [24,15] vice-versa. This will change the arising costs afterwards; the same procedures from the previous parts must be carried out until no feasible solutions exist. Also for each model run, Available budget, Overall travel time, actually used budget and Number of implemented line must be collected. The last part consists limiting the number of lines to be implemented due the organizational reasons. Limiting line constraint has to be included in the model, and then the overall travel time and Actual used budget for specified lines must be collected. The main objectives is to set line network that connect the campus efficiently while minimize the travel times for Passengers and determine the budget needed for implementing lines.

3.2 Mathematical Model

Objective Function

$$\min F = \sum_{u \in V} \sum_{l \in L} \sum_{(i,j) \in E_l} d_{ljl} \cdot x_{ulij} \quad (3.1)$$

Constraints

Flow preservation constraints:

$$\sum_{l \in L} \sum_{(i,v) \in E_l} x_{uliv} - \sum_{l \in L} \sum_{(v,j) \in E_l} x_{ulvj} = od_{uv}, \quad \forall u, v \in V, u \neq v \quad (3.2)$$

Coupling Constraints:

$$M \cdot y_l \geq x_{ulij}, \quad \forall u \in V, l \in L, (i,j) \in E_l \quad (3.3)$$

Budget Constraints

$$\sum_{l \in L} k_l \cdot y_l \leq K \quad (3.4)$$

Line Limit Constraint

$$\sum_{l \in L} y_l \leq \text{lines} \quad (3.5)$$

The main goal of used mathematical models is to determine optimal line network with implemented lines, minimize the total travel time of passengers and actual used budget. Objective function used to minimize the total travel time for all passengers across the network. Here, x_{ulij} shows the passenger flow from origin u traveling on line l via edge i,j while d_{ljl} is travel time on Line l for edge (i,j) . Flow Preservation constraints ensures that the number of passengers who start the trip in node u and finish on node v is equal to the demand od_{uv} . The difference between inbound and outbound passengers must equal to the passenger demand od_{uv} . Coupling Constraints ensures that passengers can only travel with the line l , if the line l is installed. We must consider two cases: case 1, if line l is installed, ($y_l = 1$) then $M \cdot 1 \geq x_{ulij}$ with $x_{ulij} > 0$. Case 2, if line l is not installed ($y_l = 0$), then $M \cdot 0 \geq x_{ulij}$ with $M \cdot 0 \geq x_{ulij}$ while M means a sufficient large constant number. Budget constraints ensure that the costs for installing lines do not exceed the available budget only if line l is installed $y_{l=1}$, the costs of line l are considered in budget constraint. Line limit constraints limit the number of implementing lines to maximum.

3 Line Network Planning

Overall travel time (minutes)	12,146,148
Necessary budget (= arising costs)	473,400
Number of implemented lines	60

Table 3.1: Results of the line network problem without Budget Constraint

reference budget	10% budget	Available budget	Actual used budget	Overall travel time	Number of implemented lines
473,400	47,340	426,060	423,100	12,146,148	55
473,400	47,340	378,720	376,600	12,146,148	44
473,400	47,340	331,380	327,300	12,146,148	39
473,400	47,340	284,040	282,100	12,146,148	35
473,400	47,340	236,700	235,700	12,147,020	24
473,400	47,340	189,360	188,800	12,153,434	19
473,400	47,340	142,020	141,500	12,196,148	14
473,400	47,340	94,680	93,900	12,379,476	11
473,400	47,340	47,340	No solution	No solution	No solution

Table 3.2: Results of the line network problem with Budget Constraint

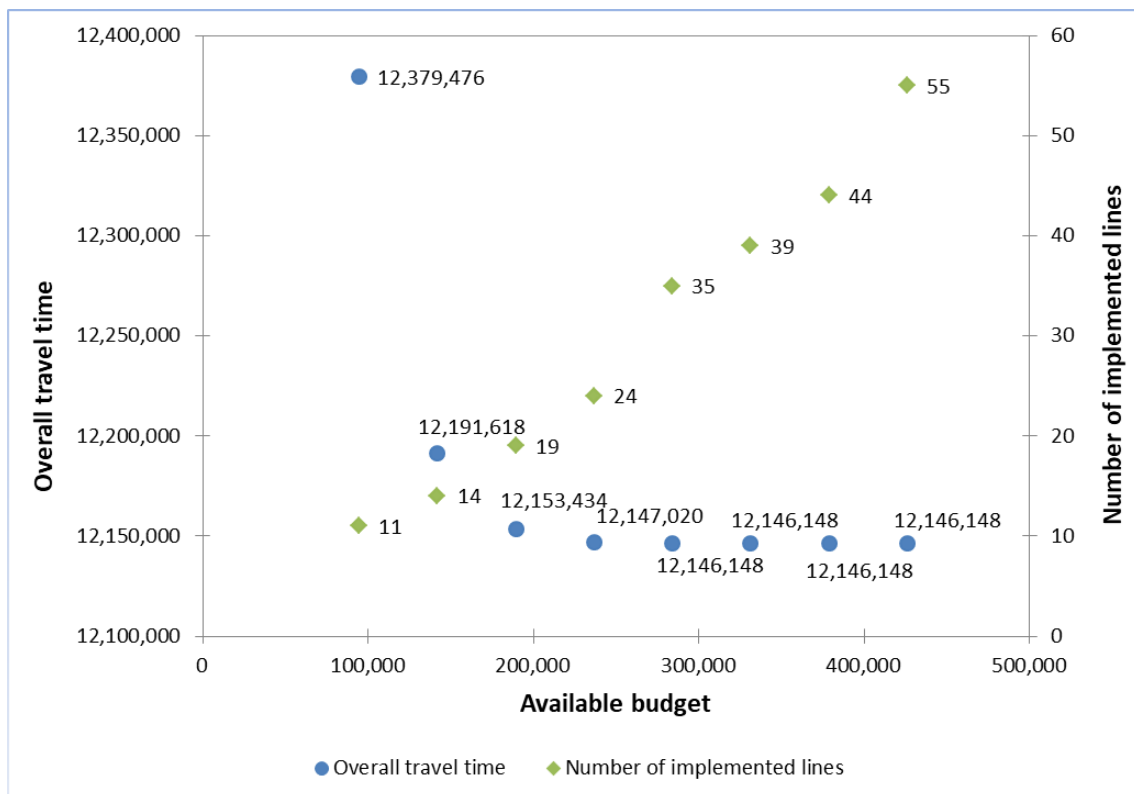


Figure 3.2: a scatter plot which shows the dependence of the travel time and the number of implemented lines from the available budget

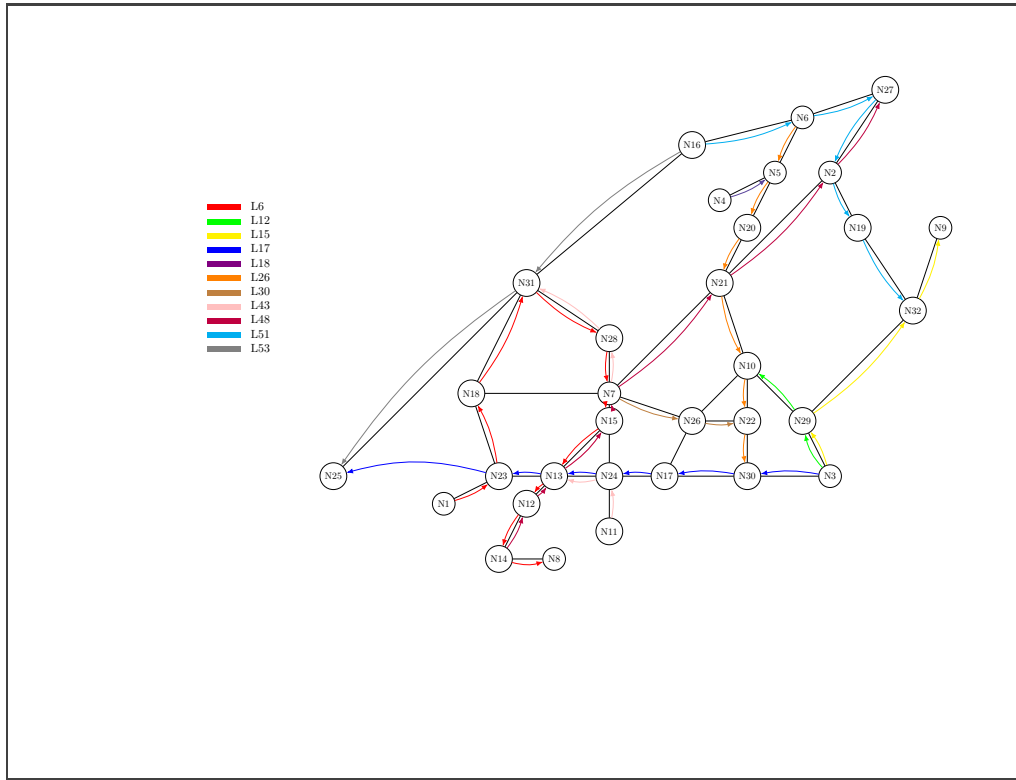


Figure 3.1: a line network for the last feasible solution

3.3 Analysis and comparison of obtained Line network results with and without budget constraint

The optimal line network without budget constraints results 3.1 demonstrates significant overall travel time, Necessary budget (arising costs) and the number of implemented lines in the network. The overall travel time for all passengers reaches 12,146,148 minutes. These Travel time is high due to that all lines were implemented and this cause the network becomes inefficiency. The total operational cost for implementing all lines is 473,400 Euro. This Amount reflects the maximum possible expenditure, as the model does not yet implement the Budget constraints. The Optimal line network with budget constraints 3.2 , iterate constantly 10% budget reduction from the budget of the arising costs of the reference network3.1. First iteration cuts to €426,060 maintain identical travel time (12,146,148 minutes) in spite of the number of implemented lines reduced to 55 lines, indicate the network flexibility to hold high demand. This travel time remains unchangeable until the reduction of the available budget to €284,040, despite the number of number implemented lines reduced to 35 lines respectively to the identified available budget. At €236,700 half of budget from reference network, line reduced to 24 causes travel time to increase to 12,147,020 minutes, indicating lack of external connectivity. As the budget continues to be reduced to €142,020(30% reference network), where only 14 lines was installed increases the travel time of 12,196,148 minutes that causes passen-

3 Line Network Planning

ger transfers becomes inefficient. The last possible solution (€94, 680, 20% of reference budget) display only 11 lines yield travel time of 12,379,476 minutes- exposing passenger transfers becomes higher inefficiency. Scatterplot 3.2 shows nonlinear declination. Travel time remains linearly until the available budget falls below 50% of reference network despite number of implemented lines was decreasing while Figure3.1 describe the implementation of line network of last possible feasible solution, each colour specify a certain line path.

Reference Budget	10% Budget	Available Budget	Actual Used Budget	Overall Travel Time (min)	Number of Implemented Lines
508,500	50,850	457,650	455,500	11,225,234	55
508,500	50,850	406,800	405,100	11,225,234	43
508,500	50,850	355,950	353,500	11,225,234	36
508,500	50,850	305,100	296,200	11,225,234	31
508,500	50,850	254,250	253,000	11,225,234	23
508,500	50,850	203,400	202,000	11,227,791	18
508,500	50,850	152,550	150,800	11,244,223	15
508,500	50,850	101,700	100,400	11,346,018	14
508,500	50,850	50,850	-	No Solution	No Solution

Table 3.3: Results of line network problem considering Diversion via Node 24)

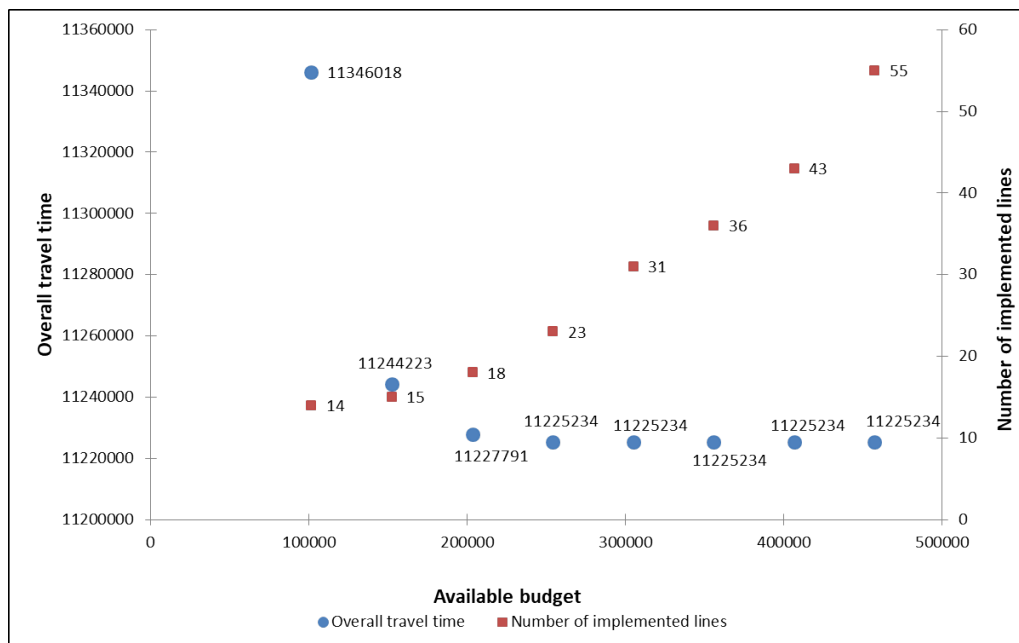


Figure 3.3: a scatter plot which shows the dependence of the travel time and the number of implemented lines from the available budget for line network problem with diversion via node 24

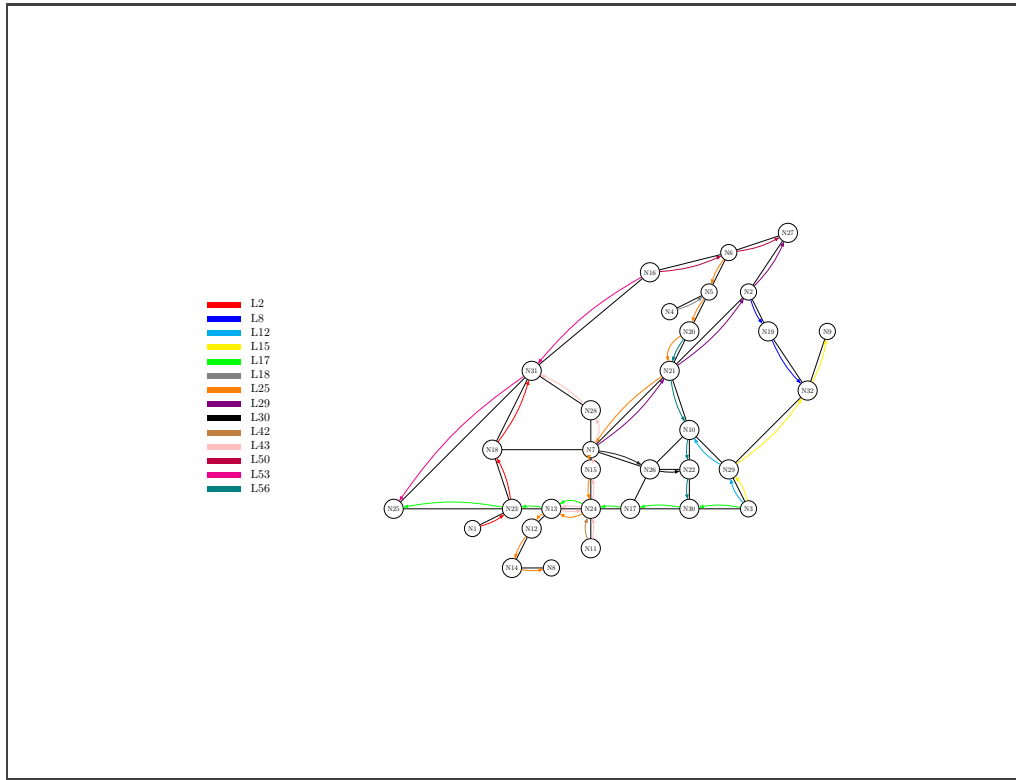


Figure 3.4: a line network for the last feasible solution for line network problem with diversion via node 24

3.4 Analysis and comparison of obtained results of line network when considering diversion of line on edge[13,15] or[15,13] via node 24.

Line network which include diversion via node 24 that means every line network within a path of edge [13,15] have to be changed to [13,24] and [24,15] vice-versa. This diversion increases the operation costs for 13 affected lines by including extra edges. The reference budget will be increased to €508,500 due to the increasing of operation costs. The same approach a constant of 10% reductions (€50,850) of the reference budget must be reduced to the current available budget. First iteration cuts to €457,650 maintain identical travel time (11,225,234 minutes) despite the number of implemented lines reduced to 55 lines , indicate the network flexibility to keep high demand by comparing the line network without considering diversion on node 24 its shows that the travel time was reduced due increasing of operation costs. This travel time remains the same until the reduction of half of the available budget to € 254250 despite the number of implemented lines was highly reduced to 23, by comparing line network without diversion via node 24 demonstrate that the lines reduced to 24 on this stage and the travel was already increased. At €203400 half from reference budget, line reduced to 18 causes

travel time to increase to 11,227,791 minutes, indicating loss of peripheral connectivity. As the budget continues to be reduced to €152,550(30% reference budget), where only 15 lines was implemented, increases the travel time of 11,244,223 minutes that causes passenger transfers becomes inefficient. The last possible solution (€101700, 20% of reference budget) gives only 14 lines that raise highly travel time to 11,346,018 minutes-exposing passenger transfers becomes higher inefficiency. Scatterplot 3.3 shows nonlinear declination. Travel time remains linearly until the available budget falls below 40% of reference budget despite number of implemented lines was decreasing while Figure 3.4 describe the implementation of line network of last possible feasible solution of actual used budget of €100400 , each colour specify a certain line path.

Number of Lines	Overall Travel Time	Actual Used Budget
12	12,229,534	129,800
11	12,263,882	119,000
10	12,315,070	104,600
9	12,452,142	101,600
8	12,655,038	87,200
7	13,431,442	84,400
6	15,743,960	70,400

Table 3.4: Line network problem considering the number of limited lines

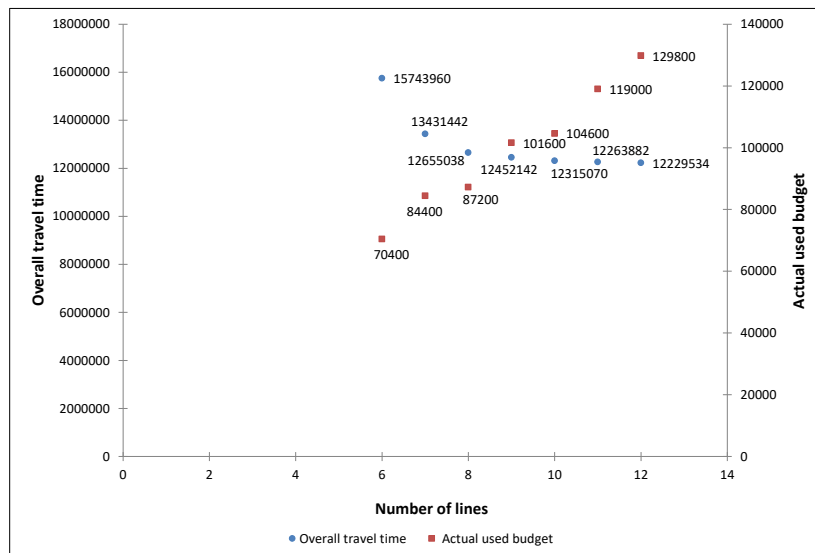


Figure 3.5: a scatter plot, which shows the dependence of the travel time and the actually used budget from the number of lines.

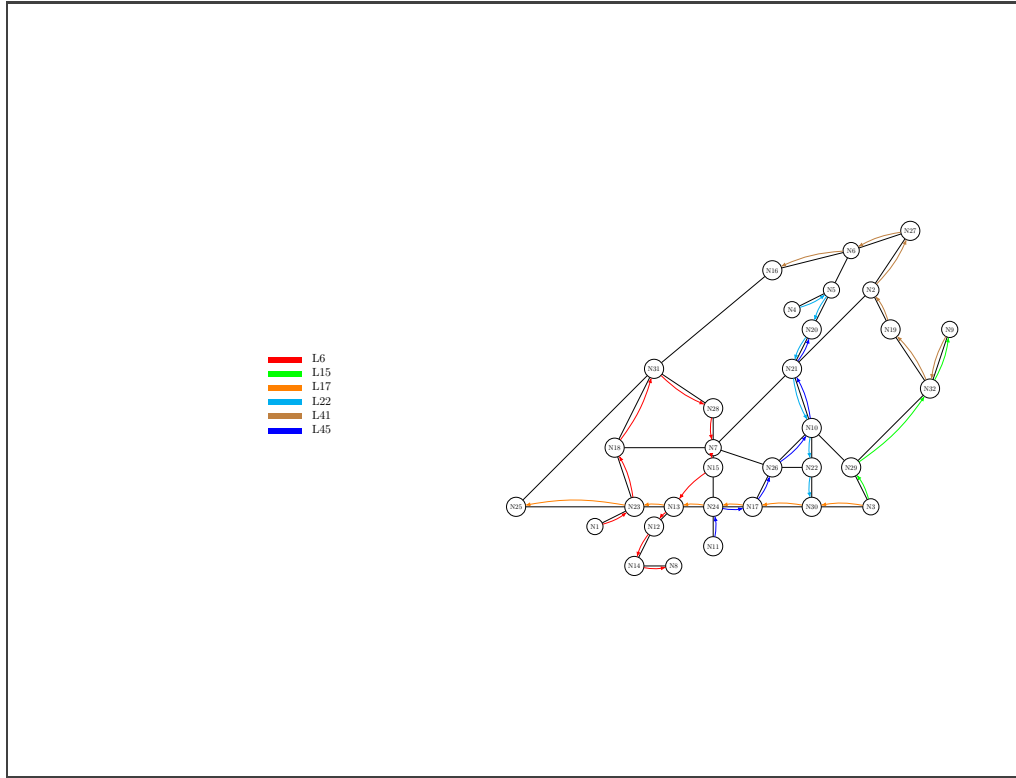


Figure 3.6: a line network for the last possible solution with the lowest number of lines

3.5 Analysis and comparison of obtained results by considering the limitation of implemented lines.

Due to the organizational reasons caused the number of lines to be limited from 12 lines and reduce the number of lines in steps 1 until no further solution provided. As the number of implemented lines reduced the overall travel times continues dramatically increasing especially for 7 and 6 lines. For the 12 lines, the overall travel time was 12,229,534 minutes with actual budget of €129,800. As the number of lines continues to decrease until 8 lines, the overall travel time slightly increase to 12,655,038 minutes. It starts to jump for 7 lines, where overall travel time of 13,431,442 minutes was obtained, which demonstrates that the passenger transfers becomes more inconvenient due to the loss of appropriate connectivity. For 6 lines, the last feasible was obtained with overall travel times 15,743,960 minutes and actual budget of €70,400, which indicates passenger inconvenience due the reduction of the direct connections and increased transfer time. From 5 lines downwards exhibits no further solution the in the model which means the line network from these lines cannot be implemented. Based on practical implementation 10 lines is highly recommended. The line network continue to work efficiently, the travel time increase by a marginal of 0.4% compared to 11 lines network, while the actual used budget decreased by 12.1%, save €14,400. Further reductions of lines experience high travel time and the budget decrease on low rate compared to 10 lines.

4 Conclusion and Future Research

This Project based on designing and implementing the optimal route network and line network by using the Mathematical Models in GAMS of campus shuttle system in Dresden. This Project was dedicated to the challenges facing TU Dresden Student and staff such as congestion and delays particularly the routes that closely link the University facilities. The main objectives were not only to design optimal route network by minimizing the costs and congestions but also to design the line network by minimizing passenger travel time. The results on route network demonstrate that large bus types (63,000 passengers/day/each directions) for 32 stops route network gave the best solution by minimizing the total costs and install few edges compared to other Bus types. adding the edge [7,10] that connect Lennéplatz and Hauptbahnhof shows that Large bus types continues to delivery efficiency solution by accumulating some demands from other edges to reduce congestion especially for the stop such as Hauptbahnhof. Afterwards, a city centre toll was introduced which cause the passenger-related usage costs increased by a factor of 8, that had changed the route network by shifting demand to the peripheral routes to reduce congestion but it had negative impact due to dramatically increasing of the usage costs to make the network work efficiently. The results on Line network by considering reference budget(€473,400) shows that travel time remains the same until the budget fell below 50% of the reference budget. For the case of diversions by replacing the edge [13,15] with [13, 24] and [24,15] Vice-versa increased the operating costs by tried to improve the network resilience, the travel time remained the same until the budget fell below 40% of the reference budget. Limiting the number implemented lines from 12 downwards showed that based on practical implementation 10 lines offers balanced the network while the least lines that can be implemented was 6. For all the cases the figures was drawn and shown the directions of passengers. To make this Campus shuttle becomes more efficiently, there are some important key areas that should be focused in future. Interpreting sensors that record the demand the route network to track the mobility of passengers whether its increasing or reducing, this will helps to get quick solution to improve the efficiency of the network. Replacing diesel buses with electrical buses would reduce emissions and save energy although it increases the infrastructure costs but this will help the city to achieve environmental sustainability goals.

Bibliography

Prof. Dr. rer. pol. habil. Jörn Schönberger : Campus Shuttle in Dresden

Statement of authorship

I hereby declare on my honor that I have prepared this project work independently. The thoughts taken directly or indirectly from external sources are marked as such. The work has not been submitted to any other examination authority and has not been published. I am aware that an untrue statement will have legal consequences.

Dresden, 17th July 2025

A handwritten signature in blue ink, consisting of stylized, overlapping loops and a long horizontal stroke extending to the right.

Olivier Nyirihirwe